

Lab 3.3.11 Report: Using Windows PowerShell

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Lab Objectives

The objective of this lab is to explore some of the functions of PowerShell including:

- Accessing PowerShell console
- Exploring Command Prompt and PowerShell commands
- Understanding cmdlets
- Using netstat command in PowerShell
- Managing Recycle Bin using PowerShell

Part 1: Access PowerShell Console

Steps Performed:

1. Opened PowerShell from Start menu
2. Opened Command Prompt from Start menu

Part 2: Explore Command Prompt and PowerShell Commands

Question 1: What are the outputs to the dir command?

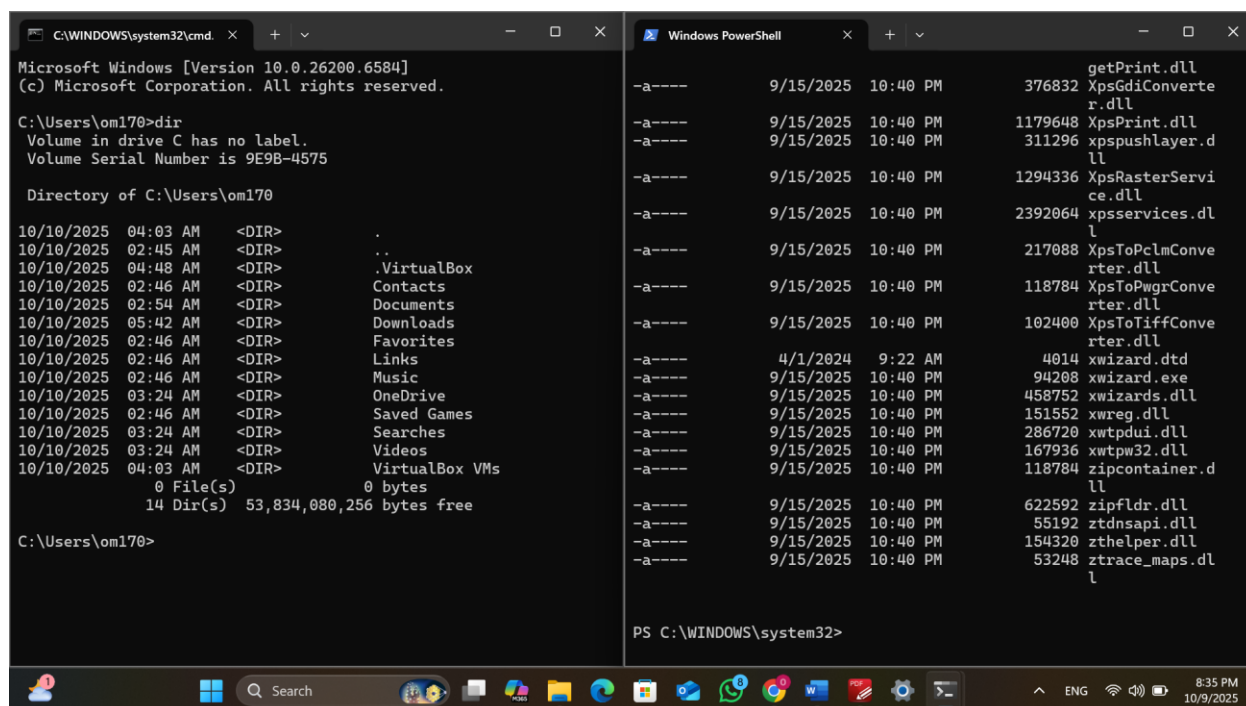
Answer:

Both Command Prompt and PowerShell display a list of files and directories in the current location. The output is similar, showing:

- Directory path
- Date and time modified
- File size
- File/folder names

However, PowerShell's output were slightly more formatted

dir Command Output:



The screenshot shows two windows side-by-side. The left window is a Command Prompt titled 'C:\WINDOWS\system32\cmd.' showing the output of the 'dir' command in the directory 'C:\Users\om170'. The output lists various folders like .VirtualBox, Contacts, Documents, Downloads, Favorites, Links, Music, OneDrive, Saved Games, Searches, Videos, and VirtualBox VMs, along with their sizes and free space. The right window is a Windows PowerShell window titled 'Windows PowerShell' showing a list of files and folders with their sizes and dates. The files listed include getPrint.dll, XpsGdiConverter.dll, XpsPrint.dll, xpspushlayer.dll, XpsRasterService.dll, xpservices.dll, XpsToPclmConverter.dll, XpsToPwgrConverter.dll, XpsToTiffConverter.dll, xwizard.dtd, xwizard.exe, xwizards.dll, xwreg.dll, xwtpdui.dll, xwtpw32.dll, zipcontainer.dll, zipfldr.dll, ztdnsapi.dll, zthelper.dll, and ztrace_maps.dll.

```
C:\WINDOWS\system32\cmd. X + -
Microsoft Windows [Version 10.0.26200.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\om170>dir
Volume in drive C has no label.
Volume Serial Number is 9E9B-4575

Directory of C:\Users\om170

10/10/2025  04:03 AM  <DIR>          .
10/10/2025  02:45 AM  <DIR>          ..
10/10/2025  04:48 AM  <DIR>          .VirtualBox
10/10/2025  02:46 AM  <DIR>          Contacts
10/10/2025  02:54 AM  <DIR>          Documents
10/10/2025  05:42 AM  <DIR>          Downloads
10/10/2025  02:46 AM  <DIR>          Favorites
10/10/2025  02:46 AM  <DIR>          Links
10/10/2025  02:46 AM  <DIR>          Music
10/10/2025  03:24 AM  <DIR>          OneDrive
10/10/2025  02:46 AM  <DIR>          Saved Games
10/10/2025  03:24 AM  <DIR>          Searches
10/10/2025  03:24 AM  <DIR>          Videos
10/10/2025  04:03 AM  <DIR>          VirtualBox VMs
0 File(s)              0 bytes
14 Dir(s)              53,834,080,256 bytes free

C:\Users\om170>

Windows PowerShell X + -
-a----             9/15/2025  10:40 PM             376832  getPrint.dll
-a----             9/15/2025  10:40 PM             1179648  XpsGdiConverter.dll
-a----             9/15/2025  10:40 PM             311296  XpsPrint.dll
-a----             9/15/2025  10:40 PM             1294336  xpspushlayer.dll
-a----             9/15/2025  10:40 PM             2392064  XpsRasterService.dll
-a----             9/15/2025  10:40 PM             217088  xpservices.dll
-a----             9/15/2025  10:40 PM             118784  XpsToPclmConverter.dll
-a----             9/15/2025  10:40 PM             102400  XpsToPwgrConverter.dll
-a----             9/15/2025  10:40 PM             4014  XpsToTiffConverter.dll
-a----             4/1/2024    9:22 AM             94208  xwizard.dtd
-a----             9/15/2025  10:40 PM             458752  xwizard.exe
-a----             9/15/2025  10:40 PM             151552  xwizards.dll
-a----             9/15/2025  10:40 PM             286720  xwreg.dll
-a----             9/15/2025  10:40 PM             167936  xwtpdui.dll
-a----             9/15/2025  10:40 PM             118784  xwtpw32.dll
-a----             9/15/2025  10:40 PM             622592  zipcontainer.dll
-a----             9/15/2025  10:40 PM             55192  zipfldr.dll
-a----             9/15/2025  10:40 PM             154320  ztdnsapi.dll
-a----             9/15/2025  10:40 PM             53248  zthelper.dll
-a----             9/15/2025  10:40 PM             53248  ztrace_maps.dll

PS C:\WINDOWS\system32>
```

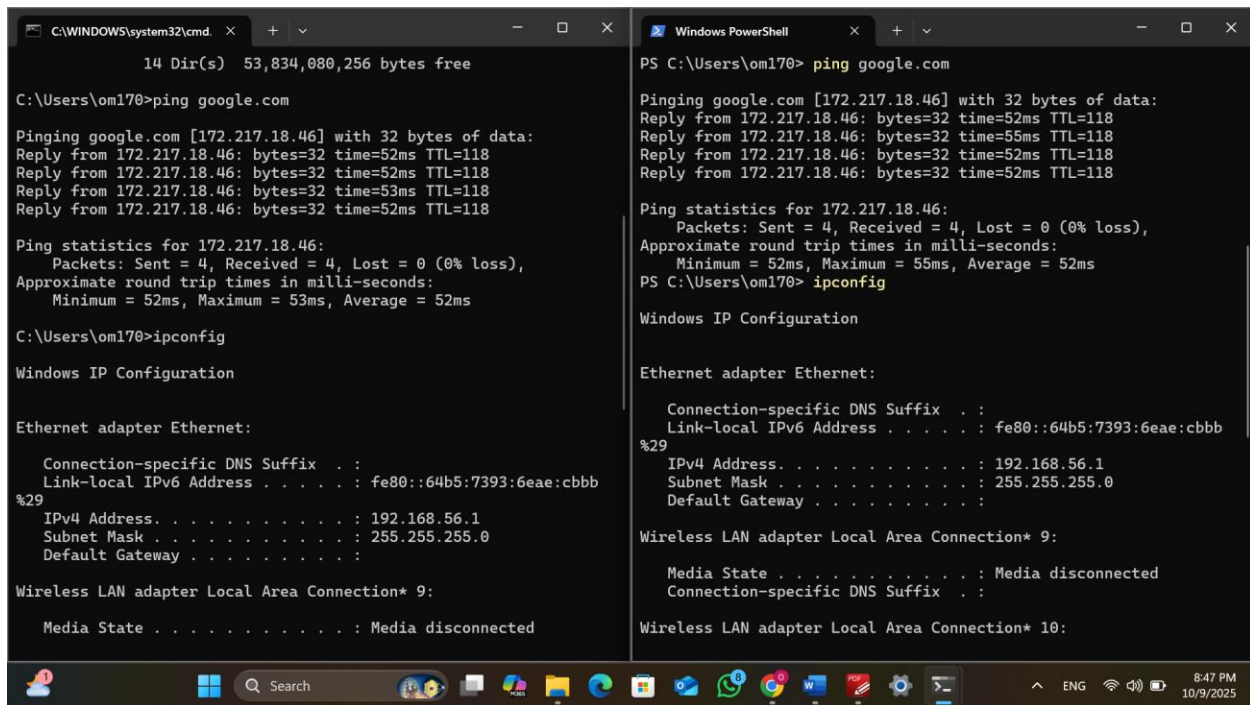
Question 2: What are the results of other commands (ping, cd, ipconfig)?

Answer:

All commands work in both Command Prompt and PowerShell:

- ping: Shows network connectivity test results to google.com
- ipconfig: Displays network configuration (IP address, subnet mask, gateway)
- cd: Changes directory successfully

Testing Multiple Commands:



The screenshot shows two side-by-side windows from a Windows 10 desktop. The left window is a Command Prompt titled 'C:\WINDOWS\system32\cmd.' showing the output of 'dir' and 'ping google.com'. The right window is a Windows PowerShell window titled 'Windows PowerShell' showing the output of 'ping google.com' and 'ipconfig'. The taskbar at the bottom shows the Start button, search bar, and various application icons. The system tray on the right shows the time as 8:47 PM on 10/9/2025.

```
C:\WINDOWS\system32\cmd. x + -
14 Dir(s) 53,834,080,256 bytes free

C:\Users\om170>ping google.com

Pinging google.com [172.217.18.46] with 32 bytes of data:
Reply from 172.217.18.46: bytes=32 time=52ms TTL=118
Reply from 172.217.18.46: bytes=32 time=52ms TTL=118
Reply from 172.217.18.46: bytes=32 time=53ms TTL=118
Reply from 172.217.18.46: bytes=32 time=52ms TTL=118

Ping statistics for 172.217.18.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 52ms, Maximum = 53ms, Average = 52ms

C:\Users\om170>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::64b5:7393:6eae:cbbb%29
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected

PS C:\Users\om170> ping google.com

Pinging google.com [172.217.18.46] with 32 bytes of data:
Reply from 172.217.18.46: bytes=32 time=52ms TTL=118
Reply from 172.217.18.46: bytes=32 time=55ms TTL=118
Reply from 172.217.18.46: bytes=32 time=52ms TTL=118
Reply from 172.217.18.46: bytes=32 time=52ms TTL=118

Ping statistics for 172.217.18.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 52ms, Maximum = 55ms, Average = 52ms
PS C:\Users\om170> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::64b5:7393:6eae:cbbb%29
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:
```

Part 3: Explore Cmdlets

Command Used:

Get-Alias dir

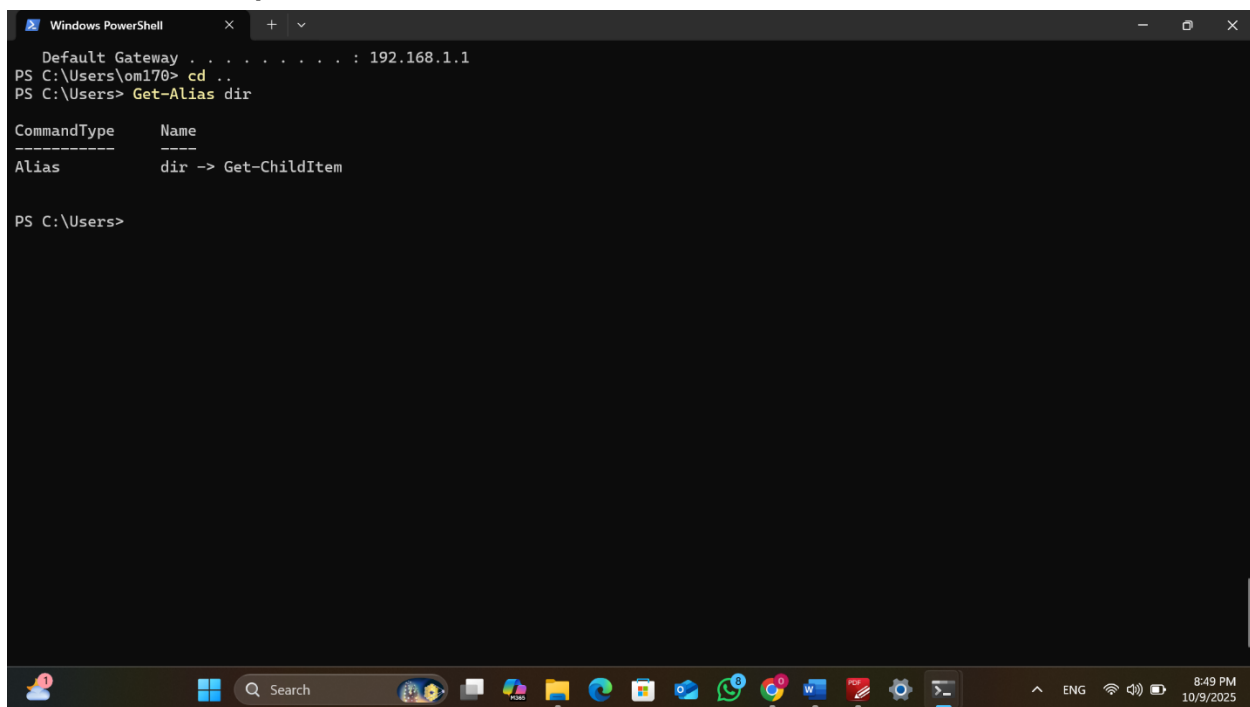
Question 3: What is the PowerShell command for dir?

Answer:

The PowerShell command for dir is: Get-ChildItem

This is a cmdlet that follows the Verb-Noun naming convention.

Get-Alias dir Output:



```
Windows PowerShell
Default Gateway . . . . . : 192.168.1.1
PS C:\Users\om170> cd ..
PS C:\Users> Get-Alias dir

CommandType      Name
-----
Alias             dir -> Get-ChildItem

PS C:\Users>
```

Part 4: Explore the netstat Command Using PowerShell

Step A: Netstat Help Command

Command Used:

netstat -h

Netstat Help Output:

```
Windows PowerShell
PS C:\Users> netstat -h

Socket Handle Count
PID          Count  Closing Count
1292          4      0
13072         1      0
4128          4      0
7208          4      0
2096          1      0
16944         7      0
12600         4      0
3132          4      0
4412          1      0
1612          10     0
1376          4      0
4452          4      0
9084          4      0
14972         6      0
9356          1      0
14504         4      0
2220          4      0
1472          4      0
1736          9      0
3820          7      0
11768         12     0
13820         3      0
PS C:\Users>
```

Step B: Display Routing Table

Command Used:

netstat -r

Question 4: What is the IPv4 gateway?

Answer:

The IPv4 gateway is: 192.168.1.1

Netstat -r Output:

```
Windows PowerShell
PS C:\Users> netstat -r

Interface List
=====
29...0a 00 27 00 00 1d .....VirtualBox Host-Only Ethernet Adapter
4...3a d5 7a 18 0a 67 .....Microsoft Wi-Fi Direct Virtual Adapter
7...3a d5 7a 18 1a 77 .....Microsoft Wi-Fi Direct Virtual Adapter #2
10...38 d5 7a 18 2a 47 .....RZ608 Wi-Fi 6E 80MHz
1.....Software Loopback Interface 1

IPv4 Route Table
=====
Active Routes:
Network Destination  Netmask          Gateway          Interface        Metric
-----
0.0.0.0              0.0.0.0          192.168.1.1      192.168.1.11     50
127.0.0.0            255.0.0.0        On-link          127.0.0.1        331
127.0.0.1            255.255.255.255  On-link          127.0.0.1        331
127.255.255.255      255.255.255.255  On-link          127.0.0.1        331
192.168.1.0          255.255.255.0    On-link          192.168.1.11     306
192.168.1.11         255.255.255.255  On-link          192.168.1.11     306
192.168.1.255        255.255.255.255  On-link          192.168.1.11     306
192.168.56.0         255.255.255.0    On-link          192.168.56.1     330
192.168.56.1         255.255.255.255  On-link          192.168.56.1     330
192.168.56.255       255.255.255.255  On-link          192.168.56.1     330
224.0.0.0            240.0.0.0        On-link          127.0.0.1        331
224.0.0.0            240.0.0.0        On-link          192.168.1.11     306
224.0.0.0            240.0.0.0        On-link          192.168.56.1     330
255.255.255.255      255.255.255.255  On-link          127.0.0.1        331
255.255.255.255      255.255.255.255  On-link          192.168.1.11     306
255.255.255.255      255.255.255.255  On-link          192.168.56.1     330

Persistent Routes:
None
```

Step C & D: Running PowerShell as Administrator

Command Used:

netstat -abno

Step E, F, G: Task Manager Analysis

Question 5: What information can you get from the Details tab and Properties dialog box for your selected PID?

Selected PID: 2220

Answer:

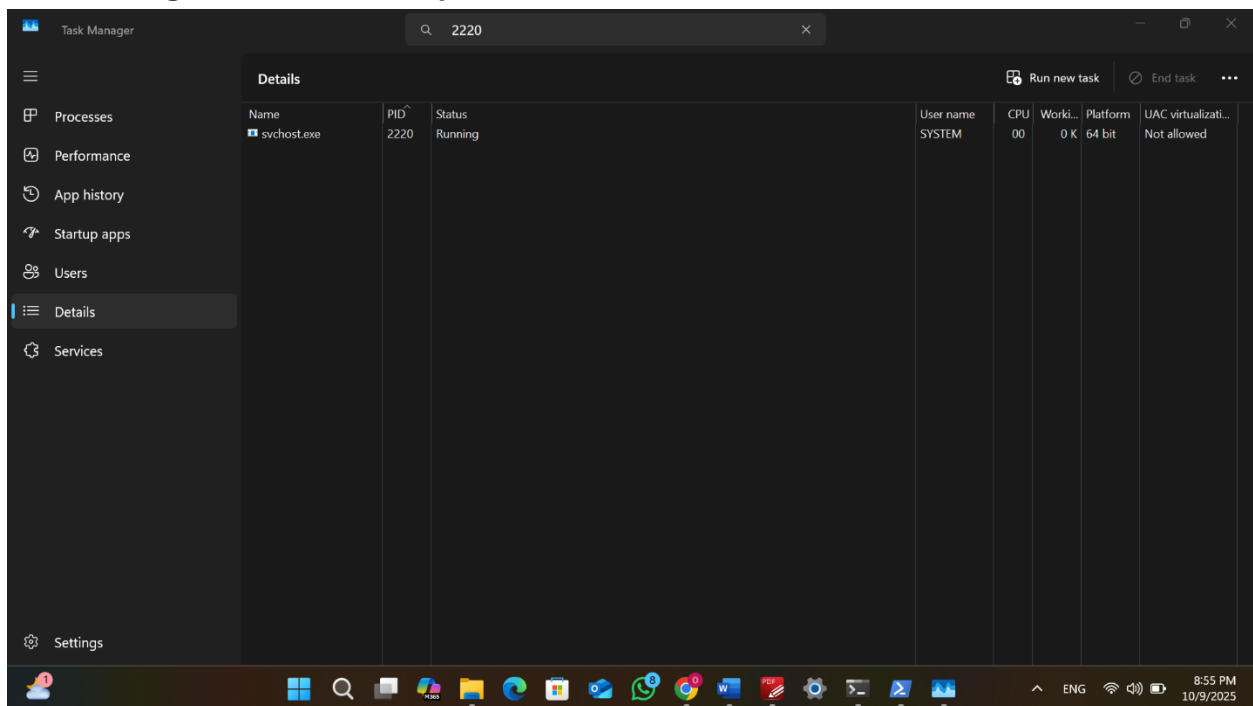
From the Details tab, I can see:

- Process name (svchost.exe)
- PID number
- Status (Running/Suspended)
- User name running the process
- CPU and Memory usage

From the Properties dialog box:

- File location
- File size
- Creation/modification dates
- Digital signature information
- Version details

Task Manager Details and Properties:



Part 5: Empty Recycle Bin Using PowerShell

Command Used:

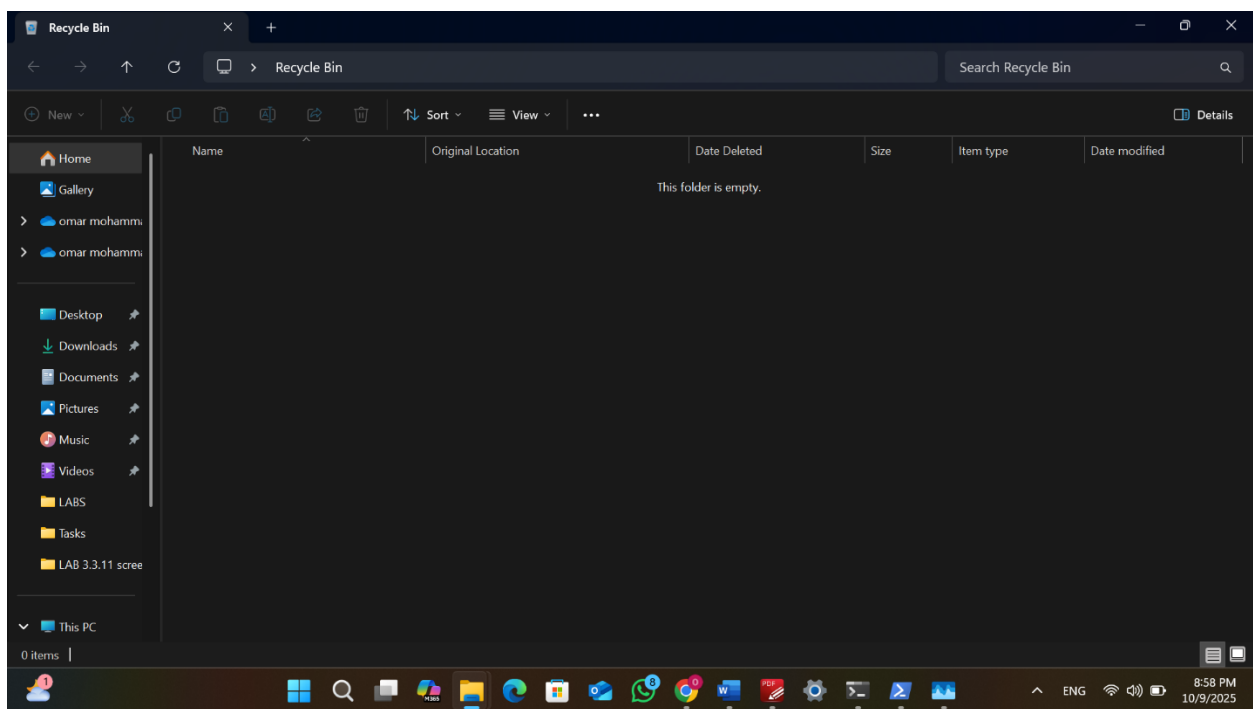
Clear-RecycleBin

Question 6: What happened to the files in the Recycle Bin?

Answer:

PowerShell asked me for confirmation, and in this case, I had to type the letter Y. Once I typed it, it immediately emptied the Recycle Bin.

Recycle Bin After:



Reflection Question

PowerShell commands for Security Analysts:

Research and record PowerShell commands that could simplify tasks as a security analyst:

My Findings:

1. Command 1: Get-EventLog

- **Purpose:** Review security and system logs for suspicious activities
- **Example:** Get-EventLog -LogName Security -Newest 100

2. **Command 2:** Get-Process

- **Purpose:** Monitor running processes and identify suspicious programs
- **Example:** Get-Process | Sort-Object CPU -Descending

3. **Command 3:** Get-NetTCPConnection

- **Purpose:** View active network connections
- **Example:** Get-NetTCPConnection | Where-Object State -eq "Established"

4. **Command 4:** Test-NetConnection

- **Purpose:** Test network connectivity and open ports
- **Example:** Test-NetConnection -ComputerName google.com -Port 443

5. **Command 5:** Get-Service

- **Purpose:** Check running services
- **Example:** Get-Service | Where-Object Status -eq "Running"

Additional Notes:

These commands help automate security monitoring, incident response, and system auditing tasks.